

§ 92. Triple equations.

The system of the nine inflection points has the property that from any two of them one obtains, by a rational process, a quite definite third one. This appears as a property of the equation of the ninth degree on which the determination of the inflection points depends, for instance of the equation whose roots are the abscissas of these points. We make the following definition.

An equation without equal roots is called a triple equation if every two of its roots determine a third, in such a way that each root of such a triple can be expressed rationally by the two others, and indeed so that in an equation

$$x_3 = \Theta(x_1, x_2) \quad (1)$$

where Θ denotes one fixed rational function, the roots of any triple may be substituted for x_1, x_2, x_3 in arbitrary order. The equation on which the inflection points depend is a triple equation of the ninth degree. There are also triple equations of other degrees, for example of the seventh degree, among which are those considered in the preceding section.¹²

We shall here consider only triple equations of the ninth degree, and first show how the notation and arrangement of the roots obtained in the preceding paragraph for the inflection points can be derived from the triple property. Each of the nine roots occurs in four triples, and hence altogether there are $4 \cdot 9 : 3 = 12$ different triples.

Take an arbitrary one of these triples and denote its roots by

$$(0, 0) \quad (1, 2) \quad (2, 1) .$$

Let $(1, 1)$ be an arbitrary fourth root. With the first three it first determines three triples, which we denote by

$$\begin{array}{lll} (1, 1) & (0, 0) & (2, 2), \\ (1, 1) & (1, 2) & (1, 0), \\ (1, 1) & (2, 1) & (0, 1), \end{array}$$

and there remain two roots which form the fourth triple with $(1, 1)$; we denote it by

$$(1, 1) \quad (2, 0) \quad (0, 2) .$$

We next seek the triple determined by $(0, 0), (2, 0)$. It cannot contain any root which occurs with either of these two roots in one of the triples already fixed; only $(0, 1)$ or $(1, 0)$ remains. Since $(0, 2)$ and $(2, 0)$ may still be interchanged, no generality is lost by taking the two further triples

$$(0, 0) \quad (2, 0) \quad (1, 0) , \quad (0, 0) \quad (0, 2) \quad (0, 1) .$$

Determining the triples which contain $(1, 0)$, one obtains

$$(2, 2) \quad (0, 1) \quad (1, 0) , \quad (1, 0) \quad (0, 2) \quad (2, 1) , \quad (0, 1) \quad (2, 0) \quad (1, 2) .$$

Finally the triples containing respectively $(0, 2)$ and $(2, 0)$ are

$$(0, 2) \quad (1, 2) \quad (2, 2) , \quad (2, 0) \quad (2, 1) \quad (2, 2) ,$$

and all twelve triples, hence the arrangement of § 91,(20), are thereby determined.

¹Triple equations of the ninth degree were first treated by Hesse in the paper "Algebraische Auflösung derjenigen Gleichung 9ten Grades etc.", Crelle's Journal, vol. 34 (1847).

²Cf. Noether, Mathematische Annalen, vol. 15 (1879), "Ueber die Gleichungen 8ten Grades und ihr Auftreten in der Theorie der Curven vierter Ordnung."

§ 93. The group of the triple equations.

The Galois group P of the triple equations is now to be determined. It is first clear that this group can contain only those permutations by which every triple again passes into a triple. For if the roots of a triple x_1, x_2, x_3 are carried by a permutation into y_1, y_2, y_3 , then, since this permutation is applicable to equation (1),

$$y_1 = \Theta(y_2, y_3),$$

and therefore y_1 is the third root of the triple determined by y_2, y_3 .

Passing to the notation

$$x = (\xi, \eta), \quad (1)$$

it follows from the arrangement of the triples that three roots

$$x_1 = (\xi_1, \eta_1), \quad x_2 = (\xi_2, \eta_2), \quad x_3 = (\xi_3, \eta_3) \quad (2)$$

form a triple if and only if

$$\begin{aligned} \xi_1 + \xi_2 + \xi_3 &\equiv 0, \\ \eta_1 + \eta_2 + \eta_3 &\equiv 0 \end{aligned} \pmod{3}. \quad (3)$$

If a permutation group of the pairs (ξ, η) , all of whose substitutions belong to the complete linear congruence group modulo 3, is called a linear group, then:

1. The group P of a triple equation is always linear.

By § 77 every permutation of the quantities (ξ, η) can in general be represented in the form

$$\xi' \equiv \varphi(\xi, \eta), \quad \eta' \equiv \psi(\xi, \eta) \pmod{3}, \quad (4)$$

where φ, ψ are integral functions with integral coefficients of ξ, η , of degree not exceeding 2 in either variable. Ordered according to η , they may be written

$$\xi' = A\eta^2 + B\eta + C, \quad \eta' = A'\eta^2 + B'\eta + C', \quad (5)$$

where A, B, C, A', B', C' are integral functions of ξ , of degree at most 2. Holding ξ fixed and putting $\eta = 0, 1, 2$, the application to the triple $(\xi, 0), (\xi, 1), (\xi, 2)$ gives that A and A' must be congruent to zero for every ξ . The same argument removes the terms in ξ^2 . Thus the substitutions have first the form

$$\begin{aligned} \xi' &\equiv m\xi\eta + a\xi + b\eta + c, \\ \eta' &\equiv m'\xi\eta + a'\xi + b'\eta + c' \end{aligned} \pmod{3}.$$

Applying these to the triple $(0, 0), (1, 1), (2, 2)$ and putting the sums of the corresponding ξ' 's and η' 's equal to zero gives $m = m' = 0$. Therefore every permutation of P is contained in

$$\begin{aligned} \xi' &\equiv a\xi + b\eta + c, \\ \eta' &\equiv a'\xi + b'\eta + c' \end{aligned} \pmod{3}. \quad (6)$$

2. Conversely, by every linear substitution of the form (6), three quantities (ξ, η) of a triple pass into three quantities (ξ', η') which likewise form a triple.

Indeed for arbitrary three pairs (6) gives

$$\begin{aligned} \xi'_1 + \xi'_2 + \xi'_3 &\equiv a(\xi_1 + \xi_2 + \xi_3) + b(\eta_1 + \eta_2 + \eta_3), \\ \eta'_1 + \eta'_2 + \eta'_3 &\equiv a'(\xi_1 + \xi_2 + \xi_3) + b'(\eta_1 + \eta_2 + \eta_3), \end{aligned} \quad (7)$$

so that the congruences (3) imply the same congruences after transformation.

It follows further that every equation of the ninth degree whose Galois group P is contained in the complete linear congruence group is a triple equation, provided the additional condition is satisfied that P carries any two assigned roots into any other two assigned roots, that is, provided

P is doubly transitive. For if P consists only of linear substitutions, then after adjunction of two arbitrary roots $x_1 = (\xi_1, \eta_1)$, $x_2 = (\xi_2, \eta_2)$, the group is reduced to one which also leaves fixed the third root $x_3 = (\xi_3, \eta_3)$ of their triple. Hence x_3 belongs to the new field of rationality and

$$x_3 = \Theta(x_1, x_2). \quad (8)$$

If P is doubly transitive, this equation is carried by permutations of P to every other ordered pair, and hence to every triple. Therefore:

3. Every equation of the ninth degree whose group is linear and doubly transitive is a triple equation.

The permutations of a doubly transitive linear group can carry every triple into every other triple, and in arbitrary order. The hypothesis of double transitivity may also be expressed by saying that the permutations of P which leave one root fixed still connect the remaining roots transitively. If P_0 denotes the subgroup which leaves a root x_0 fixed, then P_0 must still be transitive on the other roots. If P is only simply transitive, then P_0 is no longer transitive, and after adjoining x_0 the equation for the remaining eight roots splits into several factors. If P itself is transitive, x_0 may be any root, since conjugation by a permutation merely changes the notation of the roots.

Among the linear groups, the simply transitive ones have a more special character than the group of the triple equations. The Abelian group Q , consisting of

$$\xi' \equiv \xi + \alpha, \quad \eta' \equiv \eta + \beta \pmod{3}, \quad (9)$$

is only simply transitive. If P_0 is the subgroup of P which leaves $(0, 0)$ fixed, then P_0 contains only homogeneous substitutions, and is the greatest common divisor of P and the homogeneous congruence group S ; consequently $P = P_0Q$.

Using the composition series

$$S, S_1, S_2, S_3, S_4,$$

one has

$$S_4 = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \right\}.$$

Under S_4 the roots $(1, 1), (2, 2)$ only pass into each other. The group S_4Q may give every ordering of the triple $(0, 0), (1, 1), (2, 2)$, but carries it only into the two further triples $(1, 0), (2, 1), (0, 2)$ and $(0, 1), (1, 2), (2, 0)$. It is therefore only simply transitive and is also imprimitive. By adjoining

$$\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$$

one obtains S_3 ; by adjoining

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

one obtains S_2 . Thus S_2Q , and therefore also S_1Q and SQ , is doubly transitive. Proper triple equations begin only with S_2Q .

Other linear groups contained in SQ may also be formed. For instance, with the cyclic group of the third degree

$$S' = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \right\}$$

one obtains the group $S'Q = QS'$ of order 27. Under S' , $(1, 1), (2, 2)$ passes into $(1, 1), (2, 2), (1, 2), (2, 1)$, and $(1, 0), (2, 0)$, but not into $(0, 1), (0, 2)$. This group also is not the group of a triple equation. The special metacyclic equations of the ninth degree and the representation of their roots by Kronecker's principles thus still offer a field for further investigation.

§ 94. Reality relations of the triple equations.

If the field of rationality is real, then certain general conclusions can be drawn concerning the reality of the roots of a triple equation of the ninth degree. If in a triple two real roots occur, then from

$$x_3 = \Theta(x_1, x_2) = \Theta(x_2, x_1)$$

the third root must also be real. If therefore any imaginary root is present, then in each of the four triples to which it belongs at least one further imaginary root must occur. Hence the number of imaginary roots is at least five, and, since it must be even, is equal to six.

Two real roots, or two conjugate imaginary roots, always belong to a triple whose third root is real. We shall call a triple real if it contains two real roots or two conjugate imaginary roots. If imaginary roots occur in a triple, the conjugate imaginary roots also form a triple; such two triples are called conjugate imaginary triples. There are then three possibilities:

1. All roots real.
2. One real and eight imaginary roots.
3. Three real and six imaginary roots.

In the second case the four pairs of conjugate imaginary roots must determine four triples, all having the same real root as third root. If this real root is denoted by $(0, 0)$, the four conjugate pairs must be

$$(1, 2)(2, 1), \quad (1, 1)(2, 2), \quad (1, 0)(2, 0), \quad (0, 1)(0, 2).$$

In the third case the three real roots must form a triple. We take them to be

$$(0, 0) \quad (1, 1) \quad (2, 2) . \tag{1}$$

Then in the two rows

$$\begin{array}{ccc} (1, 2) & (0, 2) & (1, 0) \\ (2, 1) & (2, 0) & (0, 1) \end{array} \tag{2}$$

conjugate imaginary roots cannot occur in the same row, for otherwise the third root of the corresponding row would be real. It remains to show that the three real triples determined by the conjugate pairs must together contain all three real points. If among the three triples

$$(0, 0) \quad (1, 2) \quad (2, 1) , \quad (0, 0) \quad (2, 0) \quad (1, 0) , \quad (0, 0) \quad (0, 1) \quad (0, 2)$$

two are real, then the third is real as well; the pairs

$$(1, 2)(2, 1), \quad (2, 0)(1, 0), \quad (0, 1)(0, 2)$$

would then be conjugate pairs. But then $(1, 1), (2, 0), (0, 2)$ is a triple, whereas the conjugate roots $(1, 1), (1, 0), (0, 1)$ are not a triple, which is impossible. Therefore each of the three real roots $(0, 0), (1, 1), (2, 2)$ occurs, besides in (1), in one and only one real triple whose other two roots are conjugate imaginary. The third case therefore gives exactly the arrangement of real and imaginary roots already found for the inflection points of the curve of the third order.

That cases 1 and 2 can also occur is shown as follows. From a general equation of the ninth degree one obtains a triple equation by adjoining a function belonging to the linear group SQ . One such function is

$$\begin{aligned} V = & (0, 0)(1, 1)(2, 2) + (0, 0)(1, 2)(2, 1) + (0, 0)(1, 0)(2, 0) \\ & + (0, 0)(0, 1)(0, 2) + (1, 1)(1, 2)(1, 0) + (1, 1)(2, 1)(0, 1) \\ & + (1, 1)(0, 2)(2, 0) + (1, 2)(0, 1)(2, 0) + (2, 2)(2, 1)(2, 0) \\ & + (2, 2)(1, 2)(0, 2) + (2, 2)(0, 1)(1, 0) + (2, 1)(0, 2)(1, 0). \end{aligned}$$

This function is real in cases 1, 2, and 3. Hence the enlarged field of rationality in which the equation is a triple equation is also real.